

Qatar-1b

R-0011

Benchmark run · workflow W8-benchmark · record 2026-bench_Qatar-1_260705 · created 2026-07-08T02:48:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461226.9141 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.66 +/- 2.13 [%]
Orbital Inclination (inc)	88.39 +/- 2.85
Airmass coefficient 1 (a1)	1.94 +/- 1.06
Airmass coefficient 2 (a2)	-0.71 +/- 1.59
Scatter in the residuals of the lightcurve fit is	55.19 %
Best Comparison Star	None
Optimal Aperture	1.19
Optimal Annulus	4.75
Transit Duration (day)	0.0829 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.056 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.0829 d vs 0.0692 d (deviation 1.84σ)
Mid-time O–C	7.2 min
Depth SNR	3.3
Residual scatter	55.19 %

Light curve

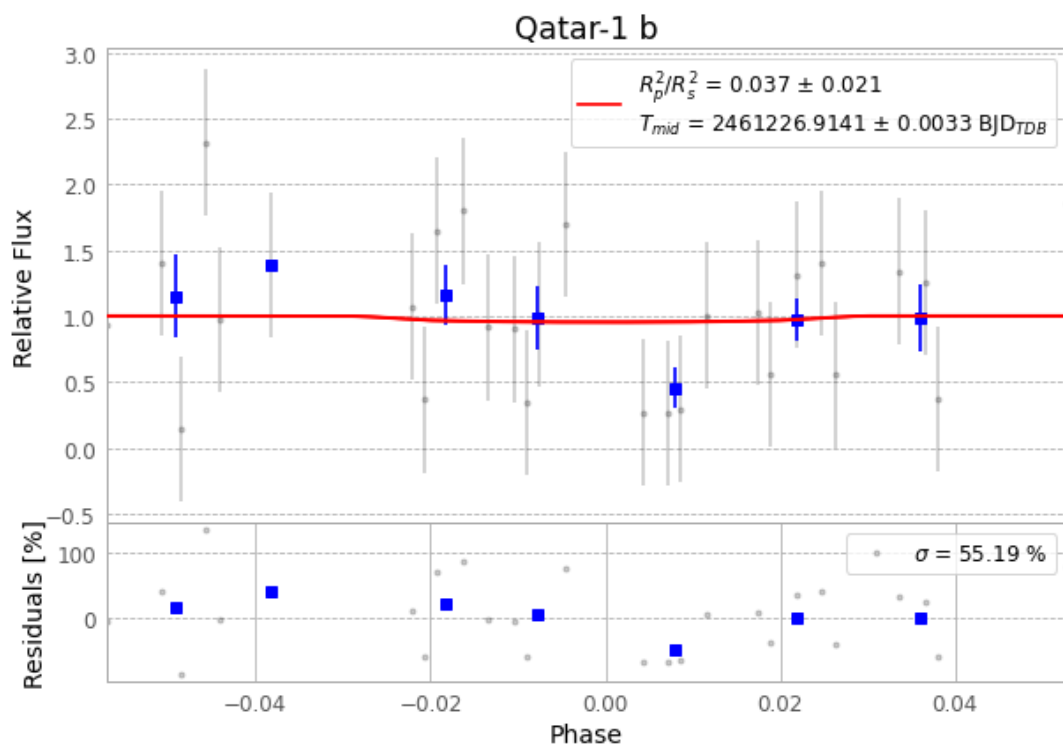


Figure 1 — FinalLightCurve_Qatar-1 b_2026-07-05.png (SHA-256 020320e6d66a32ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 102], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b88efe608f2dfe0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

75 FITS frames (49 MB), Qatar-1260705075757.FITS → Qatar-1260705120409.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260705.log (7b5e068f8e21...), FinalLightCurve_Qatar-1 b_2026-07-05.png (020320e6d66a...), FinalLightCurve_Qatar-1 b_2026-07-05.csv (15d44d3648f5...)

Compute resources

Wall time 1512.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.